

Assignment - 8

Q.1: Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x + 2x^2 \sin\left(\frac{1}{x}\right) & , \text{ if } x \neq 0 \\ 0 & , \text{ if } x = 0 \end{cases}$$

(a) By using the definition of derivative,

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{x + 2x^2 \sin\left(\frac{1}{x}\right) - 0}{x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x \left(1 + 2x \sin\left(\frac{1}{x}\right)\right)}{x} = \lim_{x \rightarrow 0} \left(1 + 2x \sin\left(\frac{1}{x}\right)\right) = 1, \text{ which exists.}$$

So, $f'(0) = 1 > 0$.

for $x \neq 0$. By using standard differentiation rules, we get

$$\begin{aligned} f'(x) &= \frac{d}{dx} [f(x)] = \frac{d}{dx} \left[x + 2x^2 \sin\left(\frac{1}{x}\right) \right] \\ &= 1 + 2x^2 \frac{d}{dx} \left[\sin\left(\frac{1}{x}\right) \right] + 2 \sin\left(\frac{1}{x}\right) \frac{d}{dx} [x^2] \end{aligned}$$

$$\therefore \underline{f'(x)} = 1 - 2 \cos\left(\frac{1}{x}\right) + 4x \sin\left(\frac{1}{x}\right), \quad x \neq 0$$

(b) We have,

$$f'(x) = 1 - 2 \cos \frac{1}{x} + 4x \sin\left(\frac{1}{x}\right), \quad x \neq 0$$

Notice the term $4x \sin\left(\frac{1}{x}\right) \rightarrow 0$ as $x \rightarrow 0$.

To make $f'(x_n) < 0$ for a sequence of points $\{x_n\}$ with $x_n \neq 0$

we need, $1 - 2\cos\left(\frac{1}{x_n}\right) + 4x_n\sin\left(\frac{1}{x_n}\right) < 0$

for small x_n ,

$$1 - 2\cos\left(\frac{1}{x_n}\right) < 0 \quad (\because 4x_n\sin\frac{1}{x_n} \approx 0)$$

So, choose sequence $x_n = \frac{1}{2n\pi + \pi/6}$ or $\frac{1}{2n\pi - \pi/6}$, $n \in \mathbb{N}$

$$\text{Then, } \cos\left(\frac{1}{x_n}\right) = \cos(2n\pi \pm \pi/6) = \sqrt{3}/2 > \frac{1}{2}$$

So, $f'(x_n) < 0$ for all sufficiently large n .

And clearly $x_n \rightarrow 0$ as $n \rightarrow \infty$

This concludes that $f'(0) = 1 > 0$ but f is not increasing near 0. ~~As~~ $\exists$ a sequence $\{x_n\} \rightarrow 0$ such that $f'(x_n) < 0$.

So derivative positivity at a point does not guarantee local monotonicity.

2. Suppose there exist a polynomial $Q(h)$ of degree $\leq r$ such that

$$\lim_{h \rightarrow 0} \frac{f(x+h) - Q(h)}{h^r} = 0$$

Define,

$$S(h) := P(h) - Q(h)$$

$$\text{Now, } f(x+h) - Q(h) = f(x+h) - P(h) + P(h) - Q(h)$$

$$\Rightarrow f(x+h) - Q(h) = R(h) + S(h)$$

By our assumption, we get

$$\lim_{h \rightarrow 0} \frac{R(h) + S(h)}{h^r} = 0 \text{ Also } \lim_{h \rightarrow 0} \frac{R(h)}{h^r} = 0$$

implies that $\lim_{h \rightarrow 0} \frac{S(h)}{h^r} = 0$.

As, $S(h)$ is a polynomial of degree less than or equal to r .

$$\text{Let } S(h) = \sum_{k=0}^r C_k h^k \text{ Then,}$$

$$\lim_{h \rightarrow 0} \frac{S(h)}{h^r} = \lim_{h \rightarrow 0} \frac{C_0 + C_1 h + C_2 h^2 + \dots + C_r h^r}{h^r}$$

$$\Rightarrow \lim_{h \rightarrow 0} \left(\frac{C_0}{h^r} + \frac{C_1}{h^{r-1}} + \dots + \frac{C_{r-1}}{h} + C_r \right)$$

For the limit to exist we have all the coefficients $C_0, C_1, \dots, C_{r-1}$ equal to zero.

$$\therefore \lim_{h \rightarrow 0} C_r = 0 \Rightarrow \boxed{C_r = 0}$$

This means all coefficients C_k are 0.

So, $S(h) \equiv 0 \Rightarrow P(h) \equiv Q(h)$. This implies that there is no such another polynomial $Q(h) \neq P(h)$ of degree $\leq r$ exists such that $\lim_{h \rightarrow 0} \frac{f(x+h) - Q(h)}{h^r} = 0$.

Therefore, $P(h)$ is unique

3. (a) By using Taylor's theorem, we know that around a , we can approximate

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a) (x-a)^k}{k!}$$

for $x = a+h$

$$\underline{f(a+h)} = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) \quad (\text{as } h \rightarrow 0 \text{ higher order term vanishes})$$

$$\text{for } x = a-h, \quad \underline{f(a-h)} = f(a) - hf'(a) + \frac{h^2}{2!} f''(a) \quad (\text{as } h \rightarrow 0 \text{ higher order terms vanish})$$

$$\therefore \lim_{h \rightarrow 0} \frac{f(a+h) - 2f(a) + f(a-h)}{h^2} \quad (\text{that involves power of } h^3)$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) + f(a-h) - f(a)}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{hf'(a)} + \frac{h^2}{2!} f''(a) - \cancel{hf'(a)} + \frac{h^2}{2!} f''(a)}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h^2} f''(a)}{\cancel{h^2}} = \underline{f''(a)}$$

(b) Let us consider the function

$$f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

Let $a = 0$.

$\therefore$ the limit of the quotient in the part (a)

$$\lim_{h \rightarrow 0} \frac{f(0+h) - 2f(0) + f(0-h)}{h^2} = \lim_{h \rightarrow 0} \frac{h^3 \sin(\frac{1}{h}) - 0 + h^3 \sin(\frac{1}{h})}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{2h^3 \sin(\frac{1}{h})}{h^2} = \lim_{h \rightarrow 0} 2h \sin \frac{1}{h} = 0$$

which exists.

By using the definition of derivative,

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^3 \sin(\frac{1}{h})}{h} = \lim_{h \rightarrow 0} h^2 \sin(\frac{1}{h}) = 0 \text{ which exists}$$

$$\therefore f'(x) = \begin{cases} 3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

Now, by using the definition of derivative,

$$f''(0) = \lim_{h \rightarrow 0} \frac{f'(0+h) - f'(0)}{h} = \lim_{h \rightarrow 0} \frac{3h^2 \sin(\frac{1}{h}) - h \cos \frac{1}{h}}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} 3h \sin(\frac{1}{h}) - \cos(\frac{1}{h}), \text{ which}$$

clearly doesn't exist since $\cos(\frac{1}{h})$ oscillates between -1 & 1 as $h \rightarrow 0$.

This shows that the limit of the quotient in part (a) exists but where as $f''(a)$ does not exist.

4. Locate and classify the point of discontinuity of the following functions f defined on $\mathbb{R}$.

$$(a) f(x) = \begin{cases} \frac{\sin x}{x} & , \text{ if } x \neq 0 \\ 0 & , \text{ if } x = 0 \end{cases}$$

We know that $\sin x$ and $\frac{1}{x}$ both are cont^s on $\mathbb{R} - \{0\}$. So, $\frac{\sin x}{x}$ the product of cont^s function is also cont^s on $\mathbb{R} - \{0\}$.

Since, $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ exists

but not equal to $f(0) = 0$. Therefore f has a removable discontinuity at $x=0$.

$$(b) f(x) = \begin{cases} e^{1/x} & , \text{ if } x \neq 0 \\ 0 & , \text{ if } x = 0 \end{cases}$$

here, $f(x)$ is a continuous function on $\mathbb{R} - \{0\}$.

Since, $\lim_{x \rightarrow 0^+} e^{1/x} = \infty$ and $\lim_{x \rightarrow 0^-} e^{1/x} = 0$

this implies at $x=0$, $\lim_{x \rightarrow 0} f(x)$ does not exist. Therefore f has an irremovable discontinuity at $x=0$.

Q.5: Let f be an increasing function on $[a, b]$ and let $x_1, x_2, \dots, x_n$ be n points on (a, b) such that $a < x_1 < x_2 < \dots < x_n < b$.

(a) Show that
$$\sum_{k=1}^n [f(x_k^+) - f(x_k^-)] \leq f(b) - f(a)$$

Sol:

Let $x_0 = a < x_1 < x_2 < \dots < x_n < x_{n+1} = b$.

For any $k = 1, 2, \dots, n$ pick any point $y_k \in (x_k, x_{k+1})$. Since, f is an increasing function on $[a, b]$ then $f(x_k^+)$ and $f(x_k^-)$ both exists for each $k = 1, 2, \dots, n$.

Because, $y_k \in (x_k, x_{k+1})$ it implies that

$$f(x_k^+) \leq f(y_k)$$

and because, $y_{k-1} \in (x_{k-1}, x_k)$ it implies that,

$$f(y_{k-1}) \leq f(x_k^-)$$

So, for each $k = 1, 2, \dots, n$, we have

$$f(x_k^+) - f(x_k^-) \leq f(y_k) - f(y_{k-1})$$

Summing up both sides from $k = 1, 2, \dots, n$, we get

$$\begin{aligned} \sum_{k=1}^n (f(x_k^+) - f(x_k^-)) &\leq \sum_{k=1}^n (f(y_k) - f(y_{k-1})) \\ &= f(y_n) - f(y_0) \quad \text{--- (i)} \end{aligned}$$

Since, $f(y_0) \geq f(x_0^+) = f(a^+)$

and, $f(y_n) \leq f(x_{n+1}^-) = f(b^-)$

$\therefore f(y_n) - f(y_0) \leq f(b^-) - f(a^+)$ - (ii)

By using eq (i) & (ii), we get

$$\sum_{k=1}^n (f(x_k^+) - f(x_k^-)) \leq f(b^-) - f(a^+) \leq f(b) - f(a)$$

($\because$ at the end points,
 $f(b^-) \leq f(b)$ & $f(a^+) \geq f(a)$)

Hence, $\sum_{k=1}^n (f(x_k^+) - f(x_k^-)) \leq f(b) - f(a)$

(b) For each $m \in \mathbb{Z}^+$, let S_m be the set of points in $[a, b]$ where the jump of f is greater than $\frac{1}{m}$. Using part (a) show that S_m is a finite set.

Sol: We know that a point x is a discontinuity of an increasing f iff the jump $J(x) := f(x^+) - f(x^-) > 0$.

Fix $m \in \mathbb{Z}^+$ and let
$$S_m = \left\{ x \in [a, b] \mid J(x) > \frac{1}{m} \right\}$$

Suppose S_m has k distinct points

$x_1, x_2, \dots, x_k$.

For each $x_i \in S_m$, with $i = 1, 2, \dots, k$

$$J(x_i) > \frac{1}{m}$$

Summing up both sides from $i = 1, 2, \dots, k$,
we get

$$\sum_{i=1}^k \frac{1}{m} < \sum_{i=1}^k J(x_i)$$

$$\Rightarrow \frac{k}{m} < \sum_{i=1}^k J(x_i) \leq f(b) - f(a) \quad (\text{By using inequality of (a)})$$

$$\Rightarrow \frac{k}{m} < f(b) - f(a)$$

Hence, $k < m(f(b) - f(a))$ which implies that
 k is bounded by the finite number $m(f(b) - f(a))$

Therefore it shows that no. of elements in
 S_m are finite.

So, S_m is a finite set.

(c) Show that the set of discontinuities
of f is countable.

Sol. Let S denotes the set of all discontinuities of f in $[a, b]$
By the observation above

$$S = \{x \in [a, b] \mid J(x) > 0\} = \bigcup_{m=1}^{\infty} \{x : J(x) > \frac{1}{m}\} \\ = \bigcup_{m=1}^{\infty} S_m$$

$\therefore$ each S_m is finite. Therefore, S is countable union of finite sets and therefore S is countable.

So, the set of discontinuities of f is countable.